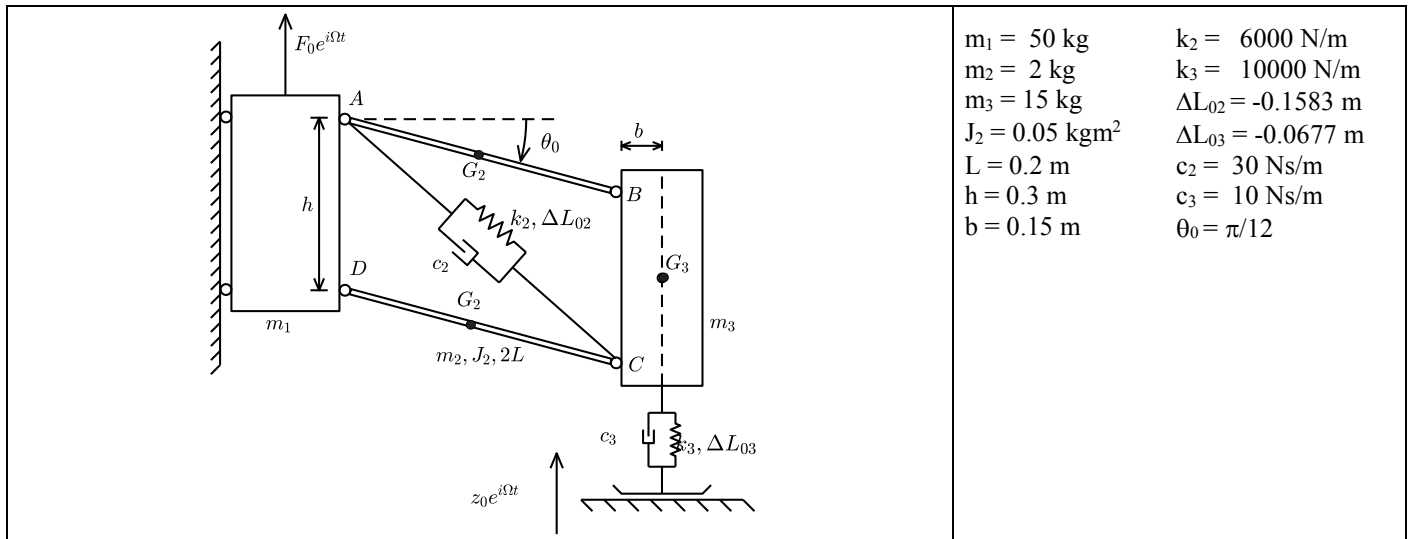


Dynamics of Mechanical Systems

November 11th, 2020



The mechanical system above lies in the vertical plane and is shown in its equilibrium position.

NOTE: Rigid bars AB and DC have the same length and inertia properties. The opposite sides of parallelogram ABCD remain parallel during the motion of the suspension.

The students are requested to:

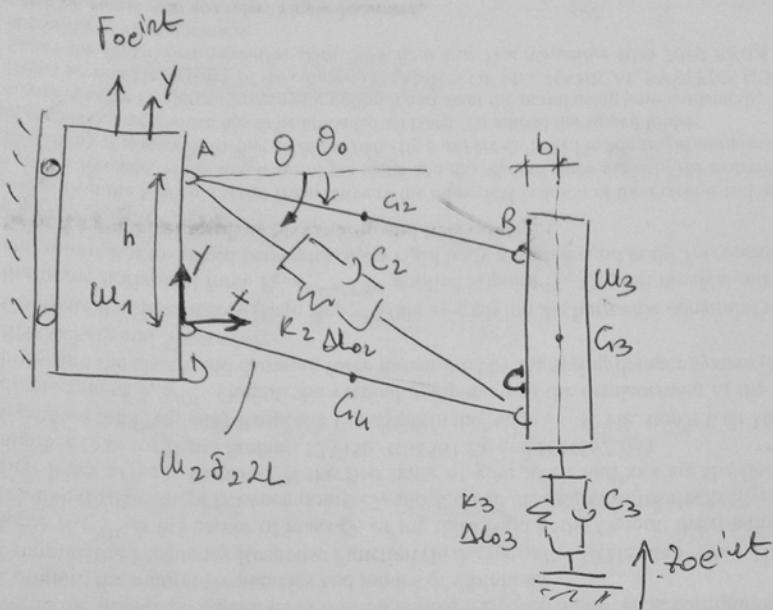
0. Write on top of each page their surname, name and ID number
1. Write the linearized equations of motion around the given equilibrium configuration; in response to this point, upload on MS-Forms one single file in pdf format containing:
 - i. the drawing of the system marked with all kinematic variables used to solve the exam;
 - ii. The vector of independent coordinates chosen to write the system's equations of motion;
 - iii. The Jacobian matrices $[\Delta m]$, $[\Delta c]$, $[\Delta k]$, with evidence of the procedure used to derive these results;
 - iv. The gravitational & second order stiffness, with evidence of the procedure used to derive these results;
2. Compute the natural frequencies and modes of vibrations;
3. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic force $F_0e^{i\Omega t}$. Output: the vertical component of the displacement of the center of mass of the rigid bar AB and its rotation. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic constraint displacement $z_0e^{i\Omega t}$. Output: the horizontal component of the displacement of the center of mass of the rigid bar AB and the reaction force in the moving constraint. Save the graphical results as SNxxx3.fig and SNxxx4.fig
5. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic constraint displacement $z_0e^{i\Omega t}$. Output: the horizontal and vertical components of the constraint force in the hinge B. Save the graphical result as SNxxx5.fig SNxxx6.fig. In response to this point, also include in the pdf file uploaded on MS-Forms:
 - v. A drawing and the symbolic equations required to derive horizontal and vertical components of the constraint force in the hinge B .

The answer to points 2 – 5 shall be provided by delivering the MATLAB script and .fig files as a compressed folder uploaded on BeeP, see instructions below.

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder **Test November 11th 2020**
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.

Written Test November 11th 2020



- 4 rigid bodies $\rightarrow$ 12 dof without constraints
 - 2 carts + 4 hinges $\rightarrow$ 10 d.o.f
- $\rightarrow$ 2 dof

$$\underline{x} = \begin{Bmatrix} \gamma \\ \theta \end{Bmatrix} \quad \underline{x}_0 = \begin{Bmatrix} \phi \\ \theta_0 \end{Bmatrix}$$

Simplified approach setting $\dot{z}_0 = 0$ and applying the equivalent force $F_{el-z} = k_3 z + C_3 \dot{z}$

$$y_A = y + h$$

$$\eta_{CH} = \eta - L \sin \theta$$

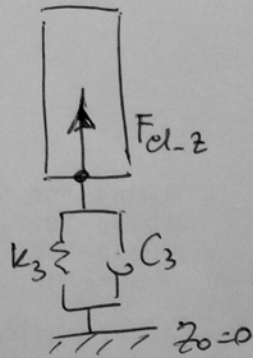
$$y_{g2} = y + h - L \sin \theta$$

$$Y_c = Y - 2L \sin \theta$$

$$x_c = 2L \cos \theta$$

$$X_{G2} = X_{G4} = L \cos \theta$$

Being ABCD an articulated parallelogram, the rotation of the AB bar is the same as that of the DC bar; the rotation of u_3 is zero.



$$T = \frac{1}{2} m_1 \dot{Y}_1^2 + \frac{1}{2} 2m_2 \dot{x}_{s4}^2 + \frac{1}{2} 2m_2 \dot{Y}_{s4}^2 + \frac{1}{2} 2J_2 \dot{\theta}^2 + \frac{1}{2} m_3 \dot{x}_{s3}^2 + \frac{1}{2} m_3 \dot{Y}_{s3}^2$$

$$D = \frac{1}{2} \sum_{i=2}^3 c_i \Delta_i^2, \quad V_{el} = \frac{1}{2} \sum_{i=2}^3 k_i \Delta_i^2;$$

$$V_g = m_1 g h_{s1} + 2m_2 g h_{s2} + m_3 g h_{s3}$$

$$\delta W = F_{o, \text{int}} \delta Y + F_{el-z} \delta Y_{s3}$$

$$[M_{PH}] = \text{diag}(m_1, 2m_2, 2m_2, 2J_2, m_3, m_3) \\ [C_{PH}] = \text{diag}(c_2, c_3) \quad [K_{PH}] = \text{diag}(k_2, k_3) \quad F = \begin{Bmatrix} F_{o, \text{int}} \\ F_{el-z} \end{Bmatrix}$$

$$\underline{Y}_u = \{Y, x_{s4}, Y_{s4}, \theta, x_{s3}, Y_{s3}\}^T$$

$$\underline{Y}_z = \underline{Y}_e = \{\Delta z, \Delta z\}^T, \quad \underline{Y}_e = \{Y, Y_{s3}\}^T$$

$$\frac{\partial x_{s4}}{\partial \theta} = -L \sin \theta, \quad \frac{\partial Y_{s4}}{\partial \theta} = -L \cos \theta$$

$$\frac{\partial x_{s3}}{\partial \theta} = -2L \sin \theta, \quad \frac{\partial Y_{s3}}{\partial \theta} = -2L \cos \theta$$

$$\Delta L_2 = L_2 - L_{\phi 2} = \sqrt{(x_c - x_A)^2 + (Y_c - Y_A)^2} - L_{\phi 2} =$$

$$= \sqrt{4L^2 \cos^2 \theta + (2L \sin \theta + h)^2} - L_{\phi 2} =$$

$$= \sqrt{4L^2 + h^2 + 4Lh \sin \theta} - L_{\phi 2}$$

$$\frac{\partial \Delta L_2}{\partial \theta} = \frac{2Lh \cos \theta}{\sqrt{4L^2 + h^2 + 4Lh \sin \theta}} \rightarrow \left. \frac{\partial \Delta L_2}{\partial \theta} \right|_{\theta_0} = \frac{2Lh \cos \theta_0}{\sqrt{4L^2 + h^2 + 4Lh \sin \theta_0}}$$

$$\Delta L_3 = Y_{s3}$$

$$[\Lambda_m] = \begin{bmatrix} 1 & 0 \\ 0 & -L \sin \theta_0 \\ 1 & -L \cos \theta_0 \\ 0 & 1 \\ 0 & -2L \sin \theta_0 \\ 1 & -2L \cos \theta_0 \end{bmatrix}$$

$$[\Lambda_k] = [\Lambda_c] = \begin{bmatrix} 0 & \left. \frac{\partial \Delta L_1}{\partial \theta} \right|_{\theta_0} \\ 1 & -2L \cos \theta_0 \end{bmatrix}$$

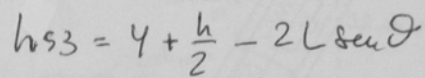
$$[\Lambda_Q] = \begin{bmatrix} 1 & 0 \\ 1 & -2L \cos \theta_0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta L_2}{\partial \theta^2} = \frac{-2Lh \sin \theta \sqrt{4L^2 + h^2 + 4Lh \sin \theta} - \frac{4L^2 h^2 \cos^2 \theta}{\sqrt{4L^2 + h^2 + 4Lh \sin \theta}}}{4L^2 + h^2 + 4Lh \sin \theta}$$

$$\frac{\partial^2 \Delta L_3}{\partial \theta^2} = 2L \sin \theta$$

$$[K_{EI, II 2}] = \begin{bmatrix} 0 & 0 \\ 0 & \left. \frac{\partial^2 \Delta L_2}{\partial \theta^2} \right|_{\theta_0} \end{bmatrix} k_2 \Delta \theta_2$$

$$[K_{EI, II 3}] = k_3 \Delta \theta_3 \begin{bmatrix} 0 & 0 \\ 0 & 2L \sin \theta_0 \end{bmatrix}$$

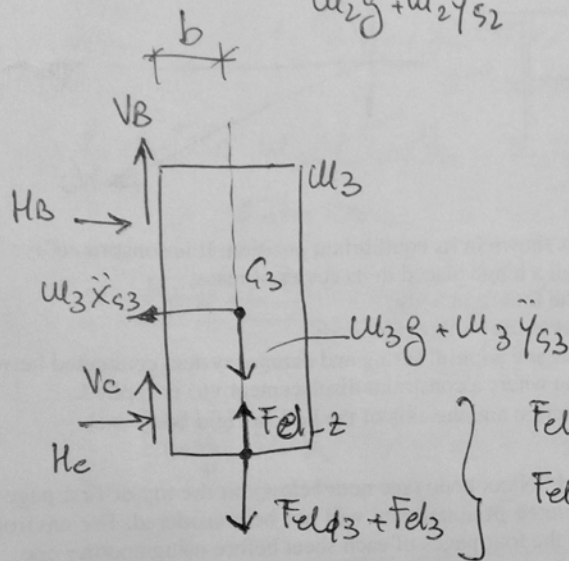
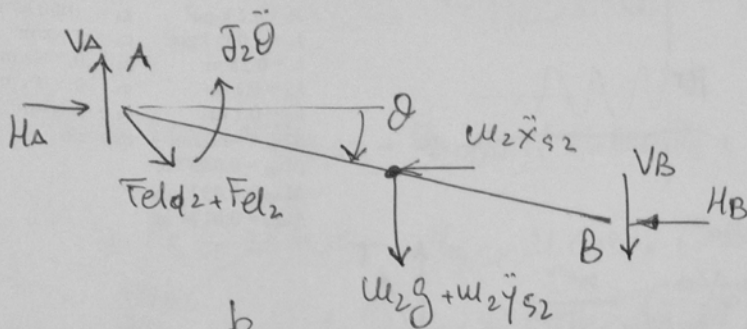


$$\frac{\partial^2 K_{52}}{\partial \theta^2} = L \sec \theta \quad ; \quad \frac{\partial^2 K_{53}}{\partial \theta^2} = 2L \sec \theta$$

$$\begin{aligned} (k_{s2}) &= 2m_2 g \begin{bmatrix} 0 & 0 \\ 0 & L \sin \theta_0 \end{bmatrix} \\ (k_{s3}) &= m_3 g \begin{bmatrix} 0 & 0 \\ 0 & 2L \sin \theta_0 \end{bmatrix} \end{aligned}$$

- Question 4: $F_2 = (k_3 + i\eta c_3)(y_{s3} - z_0)$

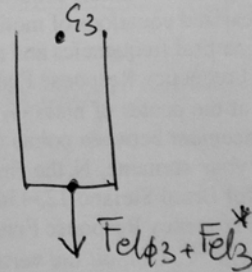
Question 5 :



$$F_{el-z} = (k_3 + i \omega c_3) z_0$$

$$F_{el3} = (k_3 + i \omega c_3) y_{G3}$$

or :



being in this case

$$F_{el3}^* = (k_3 + i \omega c_3) (y_{G3} - z_0)$$

$$\sum M_C^{u_3} = 0 :$$

$$H_B h - m_3 \ddot{x}_{G3} \frac{h}{2} + (m_3 g + m_3 \ddot{y}_{G3} + F_{el3} + F_{el3}^*) b = 0$$

$$H_{B, ST} = - \frac{(m_3 g + F_{el3}) b}{h}$$

$$H_{B, dyn} = \frac{1}{2} m_3 \ddot{x}_{G3} - (m_3 \ddot{y}_{G3} + F_{el3}^*) \frac{b}{h}$$

$$\sum M_A^{AB} = 0: \quad V_B 2L \cos \vartheta + H_B 2L \sin \vartheta + m_2 \ddot{x}_{s2} L \sin \vartheta + (m_2 g + m_2 \ddot{y}_{s2}) L \cos \vartheta - \delta_2 \ddot{\theta} = 0$$

$$H_B 2L \sin \vartheta \approx (H_{B,ST} + H_{B,dyn}) (2L \sin \vartheta_0 + 2L \cos \vartheta_0 (\vartheta - \vartheta_0)) =$$

$$= \underbrace{H_{B,ST} 2L \sin \vartheta_0}_{\text{static term}} + \underbrace{H_{B,ST} 2L \cos \vartheta_0 (\vartheta - \vartheta_0) + H_{B,dyn} 2L \sin \vartheta_0 + H_{B,dyn} 2L \cos \vartheta_0 (\vartheta - \vartheta_0)}_{2^{\text{nd}} \text{ order term}}$$

$$m_2 g \cos \vartheta \approx \underbrace{m_2 g L \cos \vartheta_0}_{\text{static term}} - m_2 g L \sin \vartheta_0 (\vartheta - \vartheta_0)$$

$$V_B = - \frac{H_{B,ST} 2L \cos \vartheta_0 (\vartheta - \vartheta_0) + H_{B,dyn} 2L \sin \vartheta_0 + m_2 \ddot{x}_{s2} L \sin \vartheta_0}{2L \cos \vartheta_0} +$$

$$+ \frac{m_2 g L \sin \vartheta_0 (\vartheta - \vartheta_0) - m_2 \ddot{y}_{s2} L \cos \vartheta_0 + \delta_2 \ddot{\theta}}{2L \cos \vartheta_0}$$

The following form, without the terms depending on the static forces, has not been considered wrong:

$$V_B = - \frac{H_{B,dyn} 2L \sin \vartheta_0 + m_2 \ddot{x}_{s2} L \sin \vartheta_0 + m_2 \ddot{y}_{s2} L \cos \vartheta_0 - \delta_2 \ddot{\theta}}{2L \cos \vartheta_0}$$